

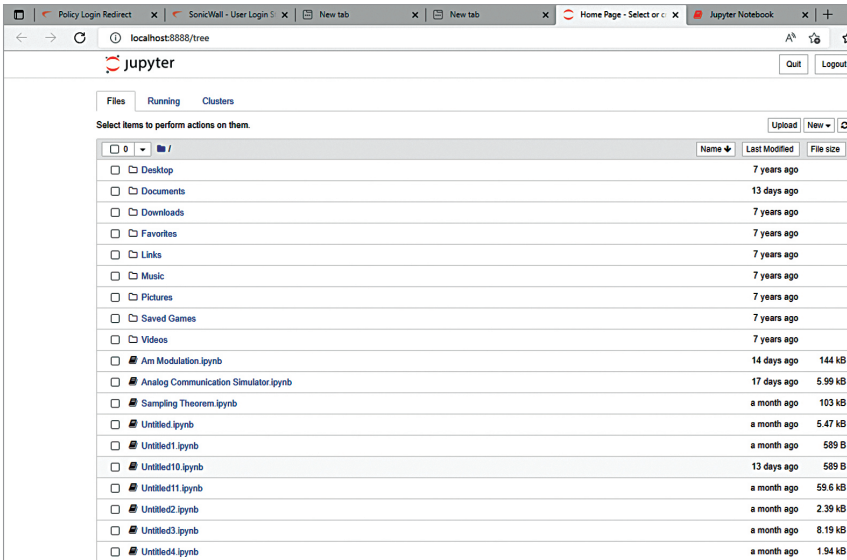


Fig. 2: Jupyter library

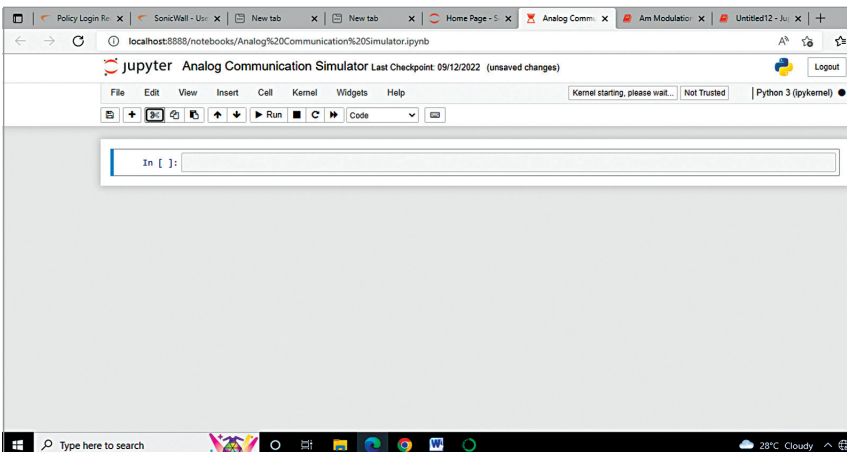


Fig. 3: Jupyter Notebook GUI

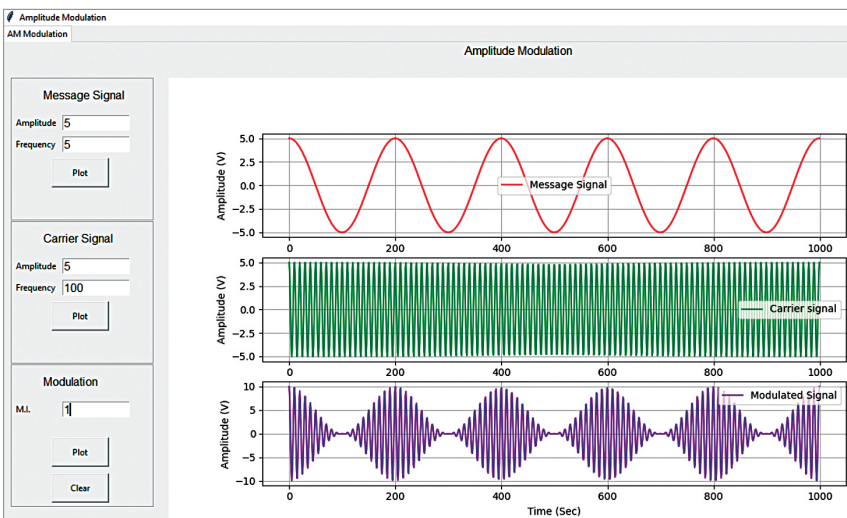


Fig. 4: GUI for simulator

with canvas to plot graph and entry boxes to get user data for amplitude, frequency of message signal and carrier signals, and for modulation index. It includes three buttons to plot message signal, carrier signal, and modulated signal, and for clearing the canvas (see Fig. 4).

In the code, first import various packages of Python for smooth operation of various functions, as follows:

```
from tkinter import *
from matplotlib.figure import Figure
from matplotlib.backends.backend_tkagg
import (FigureCanvasTkAgg, Navigation
Toolbar2Tk)
from tkinter import ttk
import numpy as np
```

To display root window and manage other components in this window, use Tk() function for first displaying the root window. The event loop of this window is created by mainloop() function. The title of root window can be provided here. Following coding snippet shows this procedure:

```
root = Tk()
root.title('Amplitude Modulation')
fig1 = Figure(figsize=(2, 2), dpi=100)
```

An interface between the Figure and Tkinter Canvas is provided by the FigureCanvasTkAgg class. Here, you may design a canvas of width = 1050 and height = 650 by positioning it at x = 230, y = 50, as follows:

```
canvas = FigureCanvasTkAgg(fig1,
master=tab1)
canvas.draw_idle()
canvas.get_tk_widget().place(x=230, y=50,
width=1050, height=650)
```

Next, create the widgets for plotting the message, carrier, and modulated signals. For user-defined values of amplitude and frequency of signal, use entry widget from the Tkinter package. These values are stored in variables for further calculation. All widgets are grouped by using frame function.